Practical Assignment: Exploring AWS → Snowflake Data Ingestion Use Cases

(Questions Only – Hands-On Exploration)

□ Use Case 1: One-Time Manual Data Load (Access Key Method)

Scenario:
A team needs to load historical CSV files from S3 into Snowflake **one time only**.

Exploration Questions

1. What Snowflake objects are mandatory to perform a one-time manual load?
2. Where and how are AWS credentials stored in Snowflake for this method?
3. What minimum S3 permissions are required for successful loading?
4. What happens if credentials are revoked after the stage is created?
5. How can you confirm which files were successfully loaded?
6. What happens if the same file is re-uploaded to S3 and reloaded?
7. How does Snowflake prevent or allow duplicate loads?
8. What security risks are visible during implementation?
9. How would this approach behave in a production environment?
10. Why is this method discouraged for long-term use?

□ Use Case 2: Secure Batch Loading (Storage Integration + COPY INTO)

Scenario:
A finance team loads daily CSV files securely from S3 using IAM roles.

Exploration Questions

1. What AWS and Snowflake components are involved before loading begins?
2. How does Snowflake assume the IAM role during loading?
3. What is the purpose of STORAGE_ALLOWED_LOCATIONS?
4. What happens if Snowflake tries to access a path outside the allowed location?
5. Can multiple stages share the same storage integration? Why?
6. How does Snowflake track loaded vs unloaded files?
7. What happens when new files are added to the same S3 folder?
8. How does this method compare to access-key loading in terms of security?
9. How can this method be automated using Snowflake Tasks?
10. What production-level advantages does this approach provide?

□ Use Case 3: Near Real-Time Ingestion (Snowpipe)

Scenario:
An application generates CSV files every few minutes and requires **automatic ingestion** into Snowflake.

Exploration Questions

1. What triggers Snowpipe to start loading files automatically?
2. Why is an event notification required in AWS?
3. Which Snowflake object listens to file arrival events?
4. Does Snowpipe require a virtual warehouse? Why?
5. How does Snowpipe ensure files are loaded only once?
6. What happens if Snowpipe is paused?
7. How are failed files handled in Snowpipe?
8. How can you monitor Snowpipe ingestion status?
9. What are the cost implications of Snowpipe vs batch loading?
10. In which scenarios is Snowpipe the best architectural choice?

□ Use Case 4: Querying Data Lake Directly (External Tables)

Scenario:
A data analytics team wants to query large historical datasets **without moving data from S3**.

Exploration Questions

1. Where does the data physically reside when using external tables?
2. What metadata does Snowflake store for external tables?
3. How does Snowflake read CSV files during query execution?
4. Does querying an external table use warehouse compute?
5. What happens if a file is deleted or modified in S3?
6. How do you refresh external table metadata?
7. Can external tables be used for transformations?
8. How does schema-on-read differ from schema-on-write?
9. What are the performance limitations of external tables?
10. When should external tables be avoided?

🟤 Cross-Method Comparison (Critical Thinking)

1. Which method provides the **highest security** and why?
2. Which method is best for **large historical data**?
3. Which method is best for **frequent small file ingestion**?
4. Which method gives the **best query performance**?
5. Which method has the **lowest operational overhead**?
6. How does cost vary between the four methods?
7. Which method aligns best with **data lake architecture**?
8. Which method would you choose for compliance-heavy environments?
9. How does failure handling differ across methods?
10. Which method would you recommend for enterprise production pipelines?